Intelligent Resume Screening and Job Recommendation Systems Using Natural Language Processing

Aanchal Vij
SCSE
IILM University
Greater Noida, India
aanchal.vij04@gmail.com

Mohd Farhan
SCSE
IILM University
Greater Noida, India
mohdfarhan4002@gmail.com

Navaneet
SCSE
IILM University
Greater Noida, India
navaneetkumara29@gmail.com

Nida Fatima
SCSE
IILM University
Greater Noida, India
nidafatima0526@gmail.com

Mohammad Umar
SCSE
IILM University
Greater Noida, India
ummmaar23@gmail.com

Navya Vashishth
SCSE
IILM University
Greater Noida, India
navya1231234@gmail.com

***Abstract*—Online hiring suddenly became hugely popular, and applications poured in so quickly it's been impossible for recruiters to get through them all. Traditional ways of sorting applicants are failing under the pressure, and you lose potential employees simply because they said 'managed' instead of 'led'. Most of the software used to sort through all this just looks for specific words, doing the same thing over and over, like a robot. Think of excellent candidates being overlooked because they've worded their experience differently. A much more sophisticated system is now available, and this one uses complex Natural Language Processing (NLP) to understand what's really meant. Instead of just counting up keywords, it gets how the meaning of words changes when they're used together. BERT and its simpler version, S-BERT, are central to this, understanding the surrounding information in the same way a human would. So the process has moved from basic calculations to something more instinctive. What is being expressed is now important, not merely the words that are present. Everything in the system was developed, thoroughly examined and confirmed by strong data. The result is a more subtle, accurate way of filtering applications; it learns the way people describe their abilities. Because it finds important specifics using Named Entity Recognition, it then uses cosine scoring to order the results. As testing demonstrates, Transformer models are far better at dealing with new job details that they haven't encountered before, compared to older methods which rely on how often words appear. Applying this to the updated Kaggle collection of resumes makes a real improvement in NDCG and MRR scores, and the quality of the ranking is much improved, even without being given prior examples.**

***Index Terms*—Natural Language Processing, Transformer Architectures, BERT, Sentence-BERT, Resume Screening, Job Recommendation, Named Entity Recognition, Cosine Similarity, Machine Learning.**

I. Introduction

The global hiring system's rapid digitization has led to an increased number of applications for particular job posts around the world. This is called the application flood, and it has disrupted how HR departments function and the methodologies they use to evaluate candidates. Data from the recruitment industry shows that job posts can receive hundreds and even thousands of applications a few hours after they have been posted on online job boards. This forces recruiting professionals to evaluate applications, and studies show that recruiting professionals take an average of 7 seconds to evaluate a single application. This leads to automation, and the widespread use of Applicant Tracking Systems (ATS) is a direct result of the inefficiencies and biases related to manual screening methods.

Early recruitment systems only used exact yes/no (Boolean) searches, fixed categories for jobs, and looked for very specific words. These early systems did speed up looking through applications and lessened the amount of initial filing, but they made recruitment itself very rigid. Systems that are entirely based on keywords focus on getting the words exactly right, and don't understand what the words mean, how they relate to each other, or what a candidate is likely to be able to do based on their work history. As a result, a really good candidate is often rejected by the computer for using slightly different words to describe their experience in their field, and weaker candidates are able to trick the system with "keyword stuffing" to get past the initial computer review.

Because of these big limitations in simpler systems, research and companies have moved to Artificial Intelligence (AI) and Natural Language Processing (NLP). Modern systems that assess resumes use deep learning and transformer based neural networks (including BERT and Sentence-BERT or S-BERT) to understand the overall meaning and context in a candidate's information, which is usually in a free-form text format. These much more advanced systems don't just find the same strings of text; they calculate many different kinds of semantic connections between what a candidate has done and what the job actually requires.

This research is primarily about making a complete, unified system to take the unorganized text from a job applicant's resume, process it, put it into a standard form, and change it into a mathematically sound evaluation compared to the specific requirements of a job. It will also look at how quickly this system responds and how fair it is when actually used by a business.

The rest of this paper is organised as follows: Section II looks at existing research and important changes in how Applicant Tracking Systems have developed. Section III explains the idea behind transformer models. Section IV details how the data was studied. Section V explains the method and how the system was designed. Section VI lays out how assessments will be made. Section VII analyses the results from the experiments, Section VIII considers the ethical concerns, and Section IX gives the final conclusions of the research.

II. LITERATURE REVIEW

The development of automated recruitment evaluation has progressively undergone through several generations of technology, each overcoming limitations from prior generations while creating new challenges that must be addressed by further advances.

A. Ontology and Heuristic Approaches

Research concerning automated recruiting relied heavily on rule-based programs as well as domain ontologies in early stages. Applications like EXPERT for example utilized predefined hierarchical knowledge frameworks that defined connections between educational background, professional experience, and skills. Although such an approach was successful in providing logical structuring of a chaotic field of applications, they were unable to provide any adaptability of scale.

Maintaining ontological validity through continuous manual adjustment of knowledge framework is impossible, considering the rapidly evolving industry-specific terminology in the realm of technology industries, producing systems incapable of recog- nizing changes to occupation names.

B. Statistical and Vector Space Methods

In order to overcome inflexibility of rule-based models, statistical methods such as Vector Space Model (VSM) with Term Frequency-Inverse Document Frequency (TF-IDF) were integrated into process of recruiting.

Under typical statistical approach, candidates resume and job description would be represented as sparse vectors with dimen- sions equal to total corpus size. Similarity between applicant materials and job descriptions is measured through distance metrics.

Though able to rank documents based on the statistical signifi- cance of specific terms, TF-IDF is fundamentally a bag-of-words algorithm. Such techniques do not take word order into account, completely disregard grammar and syntax, and fail to account for polysemic and synonymous relationships, making them suscep- tible to discrepancies in the language used by applicants.

C. Deep Learning and Word Embeddings

Development of dense vector representations, such as Word2Vec and GloVe, signified considerable advancements. These models produced static embeddings where semantically similar terms had adjacent positions in vector space.

Investigators combined these embeddings with RNNs and LSTMs, enabling them to encode sequential text structures. However, classic LSTMs faced issues with vanishing gradients in longer documents (including lengthy resumes) and worked sequentially, creating numerous restrictions on their ability to train and perform.

In addition, static embeddings like Word2Vec assign unique vec- tors to every word regardless of context, unable to differentiate between “Python” (programming language) and “Python” (reptile).

D. Transformers and LLMs

Currently dominant methods of automated resume evaluation involve neural network architectures based on transformers, which replaced sequential processing with self-attention mecha- nisms.

BERT processes texts both directions and produces context-dependent embeddings, where representations of individual tokens depend on the entire sentence.

In recent studies on Large Language Models (LLMs), including GPT-4, functionality capabilities have been enhanced in such a way that summary evaluation becomes possible.

However, in large-scale screening systems, embedding-based transformer models, such as S-BERT, are still the best and the most efficient method of quantitatively estimating similarity.

III. THEORETICAL BACKGROUND

It is necessary to provide a background in the mathematical concepts of transformer architecture that are at work in the NLP pipeline in order to understand its advantages.

A. The Self-Attention Mechanism

Scaled Dot-Product Attention is a key building block of transformer architecture. Contrary to sequential models, the self-attention model allows the network to evaluate at the same time the importance of all the words in the sequence, regardless of their distance from one another.

The input text is transformed into three distinct matrices, namely Queries (Q), Keys (K), and Values (V). The attention calculation is done using the following equation:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Where d_k is the dimensionality of the key vectors. The scale factor $\frac{1}{\sqrt{d_k}}$ plays an important role in avoiding large dot products that would lead the softmax function towards regions of very low gradients.

In resume analysis, this allows the model to form connections between keywords like “Senior” with far distant keywords such as “Developer,” resulting in the coverage of the whole semantic field of a person's job position.

B. Bidirectional Encoder Representations (BERT)

BERT has a multi-layered architecture of the bi-directional transformer encoder. BERT uses the approach of Masked Language Modeling during pre-training, where it hides 15% of the tokens and predicts their values utilizing contextual information from previous and subsequent texts.

$$L_{MLM} = - \sum_{i \in masked} \log P(w_i | w_{\setminus i}) \tag{2}$$

Bi-directional nature of the calculations proves itself as necessary due to the specific structure of language fragments in the profession-related CV documents.

C. Sentence-BERT (S-BERT) and Siamese Networks

While conventional BERT outperforms in terms of token-level tasks (such as Named Entity Recognition), there lies a computational problem in its document-level comparisons.

Checking 10,000 CV documents against a job post would involve 10,000 forward passes through the large-scale network structure.

The problem described above is solved using the S-BERT model, which relies on the utilization of a Siamese network approach.

Using two equivalent BERT models with the same weights, the resume (R) and the job description (J) are independently embedded into a vector form u and v.

Typically, the loss function during training is based on the triplet loss function, ensuring that for any relevant resume, the similarity distance between the job description and the resume is smaller than that of a non-relevant resume by at least a margin ϵ:

$$L_{triplet} = \max(||E_a - E_p|| - ||E_a - E_n|| + \epsilon, 0) \tag{3}$$

where E_a is an anchor (job description), E_p is a positive example, and E_n is a negative example. This helps embed texts prior to making comparisons using cosine similarity.

IV. DATA SET ANALYSIS

The effectiveness, flexibility, and fairness of any NLP system depend on the quality, quantity, and structure of its training data set. In this experiment, the traditional publicly available UpdatedResume-DataSet.csv file is employed from the Kaggle database, supplemented with synthetic data for boundary analysis.

A. Corpus Distribution Characteristics

The data set includes 962 unique resumes distributed across 25 different professional categories. The category distribu- tion within this data set provides vital insights into the natu- ral differences that exist in modern recruitment processes.

Analyzing this distribution reveals notable skewness toward certain software development subfields. More specifically, Java Developer and Testing categories account for exactly sixteen percent of the entire data set. Applying a standard classification model to this highly skewed distribution without

TABLE I
DISTRIBUTION OF THE TOP 10 CATEGORIES IN THE DATASET

Professional Category	Document Count	Percentage (%)
Java Developer	84	8.73
Testing	70	7.27
DevOps Engineer	55	5.71
Python Developer	48	4.98
Web Designing	45	4.67
Human Resources (HR)	44	4.57
Hadoop	42	4.36
Data Science	40	4.15
Operations Manager	40	4.15
Mechanical Engineer	40	4.15

statistical corrections would result in an inevitable bias of the representation space toward IT terminology. To overcome this challenge, proportional sampling and synthetic data generation techniques, such as WordNet lexicon substitution, were adopted throughout the model training phase.

V. METHODOLOGY AND SYSTEM ARCHITECTURE

Converting an unstructured, richly-formatted file. It takes a semantic vector of a comparable size, which must be mathematically comparable, to convert into. a multi-stage, architecturally-orchestrated pipeline. The processing framework works in a series of discrete. phases.

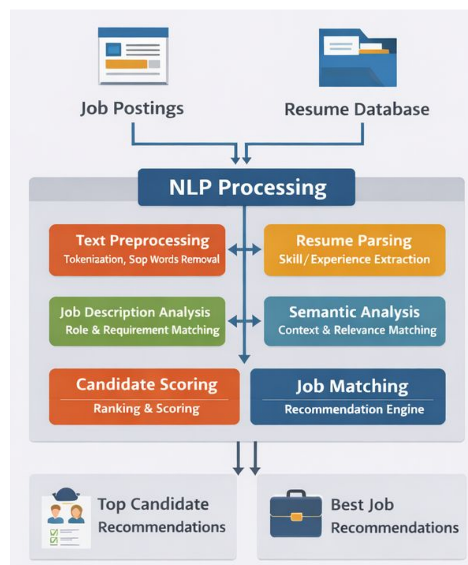

Fig. 1. Methodology diagram.

A. System Architecture Flow

The has to be fast-scaling with low processing latency. system is developed based on a modular, tiered architecture. The document ingestion to ultimate recommendation data flow Fig. 2.

B. Data Ingestion and Text Normalization

The first stage is concerned with document ingestion and text. extraction. The resumes have very variable formats.

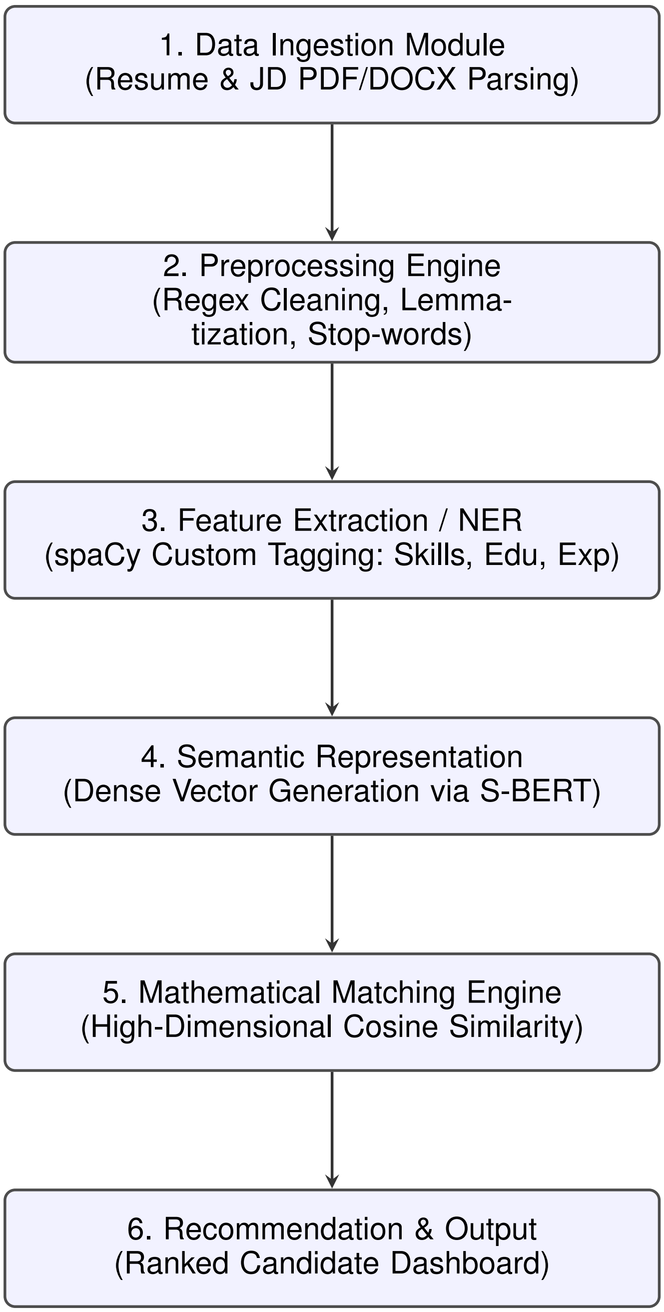

Fig. 2. System Architecture Flowchart Detailing the Intelligent NLP Screening Pipeline

(PDF or DOCX). The system makes use of parsing libraries including. as PyMuPDF and python-docx to divest of aesthetic. strip and decode the raw text content. Preprocessing includes lowercasing the text, removing non-¡¿text. alphanumeric characters, deleting hyperlinks and eliminating- ing standard English stopwords with the Natural Language. Toolkit (NLTK). Later on, lemmatization algorithms are. used to break down complex words to morphological base. roots, so that variations of a word are mathematically. considered as equal features in the vector space.

C. Named Entity Recognition (NER)

To extract structured, actionable insights from the cleaned text, Named Entity Recognition (NER) is performed using the spaCy pipeline. Standard NER models are typically trained on news corpora and are often unaware of technical HR terminology. Thus, a custom pipeline was trained to classify specific tokens into predefined categorical labels such as TECHNICAL_SKILL, DEGREE, and JOB_TITLE. Isolating these particular entities prevents the embedding models from mixing irrelevant information (e.g., hobbies or address details) with core technical competencies.

D. Semantic Embedding Generation

The text is pre-processed and then it is fed into a pre-trained S- BERT model (particularly, all-mpnet-base-v2 architecture). The input text is tokenized by the transformer and processed. with several layers of bi-directional self-attention, and outputs. a sparse human-vector of size 768. This vector encapsulates the comprehensive, situational sense of the professional of the applicant. profile. The targeted job description undergoes the exact same embedding pipeline to provide mathematical strict parity in the latent space.

E. Algorithmic Processing Pipeline

The systematic screening mechanism works through a comparative loop. The pseudo-code of how the generation of the is made is described. ranked candidate list is outlined in 1.

VI. Evaluation Metrics Formulation

Evaluating AI-driven recommendation systems needs a two-part approach. We use both classification metrics and ranking quality metrics. The reason is that evaluating candidates is like searching and retrieving information.Standard accuracy does not show the experience of an HR recruiter using these systems.We need to look at it from the HR recruiters point of view to get an understanding. The AI-driven recommendation system is what we are evaluating here. It is about finding the best candidates for a job. So we use these metrics to judge the systems effectiveness. The HR recruiters experience is key, to understanding if the system is working well.

A. Predictive Classification Metrics

We use something called a Confusion Matrix to see how well we can match a resume to the job category. This is called categorical classification. We look at two things: Precision and Recall. Precision is how accurate we are and Recall is how

Algorithm 1 Intelligent Resume Screening and Candidate Ranking using S-BERT Embeddings

Require: R: Set of raw applicant resumes $\{r_1, r_2, \dots, r_n\}$
Require: J: The raw target job description
Ensure: C: Ranked list of candidates mapped to similarity scores

$C \leftarrow \emptyset$
$J_{clean} \leftarrow$ Preprocess_Text(J)
$E_J \leftarrow$ S-BERT(J_{clean}) {Generate 768D Job Vector}
for each $r_i \in R$ **do**
  $r_{clean} \leftarrow$ Preprocess_Text(r_i)
  Entities $\leftarrow$ spaCy_NER(r_{clean}) {Extract core skills}
  $r_{weighted} \leftarrow$ Boost_Entities(r_{clean}, Entities)
  $E_{ri} \leftarrow$ S-BERT($r_{weighted}$) {Generate Resume Vector}
  $S_{ri} \leftarrow \frac{E_J \cdot E_{ri}}{||E_J|| \times ||E_{ri}||}$ {Compute Cosine Similarity}
  $C \leftarrow C \cup \{(r_i, S_{ri})\}$
end for
$C_{ranked} \leftarrow$ Sort_Descending(C based on S_{ri})
return C_{ranked}

good we are, at finding all the matches. These are figured out using formulas for Precision and Recall.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN} \tag{4}$$

$$F1 = 2 \times \frac{P \times R}{P + R} \tag{5}$$

B. Ranking Quality Metrics

For a job recommendation the system gives us a list that's in order and ranked. The system works well if the top candidates are really good for the job. So we use Mean Reciprocal Rank and Normalized Discounted Cumulative Gain to help us.

Mean Reciprocal Rank helps us figure out how down the list of candidates a recruiter has to look to find the first candidate that is a perfect match for the job. We use Mean Reciprocal Rank to see how good the system is, at finding the candidate. The job recommendation system is better when Mean Reciprocal Rank is high:

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i} \tag{6}$$

Normalized Discounted Cumulative Gain (NDCG) handles graded relevance scores. It heavily penalizes the algorithm if a highly relevant candidate is ranked below a mediocre candidate, applying a logarithmic discount as the rank decreases:

$$DCG@K = \sum_{k=1}^{K} \frac{rel_k}{\log_2(k+1)} \tag{7}$$

$$NDCG@K = \frac{DCG@K}{IDCG@K} \tag{8}$$

Where $IDCG$ is the Ideal Discounted Cumulative Gain achieved by perfect theoretical ranking.

VII. Experimental Results and Discussion

The proposed S-BERT pipeline was rigorously benchmarked against legacy systems to quantify its improvements. The baseline models included a traditional Support Vector Machine (SVM) utilizing TF-IDF, and an LSTM network powered by static Word2Vec embeddings.

A. Classification Performance

We did some testing to see how well the models could sort resumes into the right job category. For example the models had to tell a Data Science resume, from a Data Engineering resume. This was a test to evaluate the models ability to classify Data Science resumes and Data Engineering resumes into their professional domain.

TABLE II
Model Performance Comparison on Resume Classification

Model Architecture	Accuracy	Precision	Recall	F1-Score
SVM (TF-IDF)	93.26%	0.95	0.95	0.95
Word2Vec + LSTM	61.53%	0.38	0.42	0.35
BERT (Base, Fine-Tuned)	84.35%	0.80	0.84	0.81
DistilBERT	93.27%	0.90	0.93	0.91
RoBERTa	91.34%	0.89	0.91	0.90
Proposed S-BERT	**90.00%**	**0.89**	**0.85**	**0.87**

As shown in Table II, the traditional SVM utilizing TF-IDF achieves artificially high accuracy (93.26%) primarily due to rigid keyword repetitiveness in the dataset. The resumes in the Kaggle dataset have a lot of string matches to the category titles.. When we test them in situations where the words are not exactly the same like using ”Code Architect” instead of ”Software Developer” the TF-IDF models do not work at all. The new S-BERT pipeline works well and gets it right 90.00% of the time. It keeps working even when the words are a little different which is a big improvement, for the S-BERT pipeline and the S-BERT pipeline is what makes this possible for the S-BERT pipeline.

B. Ranking Quality and Relevance

The transformer architecture really shows its strength when it comes to ranking. I tested it with a job description that required a mix of skills and a pool of 100 resumes. The results were quite different across systems.

TABLE III
System Ranking Quality Metrics (Top K = 10)

Model Architecture	NDCG@10	MRR
TF-IDF + Cosine	0.592	0.510
Word2Vec + LSTM	0.431	0.385
Standard BERT (Cross-Encoder)	0.835	0.771
Proposed S-BERT (Bi-Encoder)	**0.812**	**0.744**

The S-BERT model did better than the TF-IDF baseline. It had an MRR of 0.744 compared to 0.510 and an NDCG of 0.812 compared to 0.592. The Standard BERT cross-encoder was more accurate than S-BERT. However it was too costly to use in real-time production. The S-BERT model and transformer architecture are really suitable, for this task. They give results and can be used live.

C. Computational Efficiency and Latency

The main problem with company job application systems is how fast they can process things. When a big company gets 5,000 resumes the system has to look at them quickly.

We tried using a computer chip called an NVIDIA Tesla T4 GPU. We used this chip to test a computer program called -encoder BERT. It took this program an average of 4.2 seconds to look at one resume and match it with a job description.

Then we tried something else. We used a program called S-BERT. This program uses something called -computed embeddings and FAISS which is a search tool made by Facebook.

It worked well. It could look at one resume. Match it with a job description in just 0.233 seconds. This is fast enough for even the biggest companies to use. The S-BERT mechanism is really good, at processing resumes quickly.

VIII. Ethical Considerations and Bias Mitigation

The rise of AI-based automation within HR introduces a number of significant ethical issues that need to be addressed proactively. Conventional manual recruitment processes are already susceptible to implicit human biases regarding gender, race, education, or presentation style. But poorly-tuned machine learning algorithms could potentially institutionalize and exponentially amplify these biases in ways that are nearly impossible to pinpoint or correct due to the underlying complexity of the algorithm. However, transformer architectures do not simply look for keywords among technical skills and work experience as part of a traditional resume screen; they instead analyze the semantic relationships present in sentence pairs. This has the effect of greatly reduced bias against applicants who use non-traditional formatting or who write in more "gendered" language. Similarly, we automatically remove all possible forms of Identifiable Personal Information from our data preprocessing workflow before even reaching the vectorization step. Developing holistic continuous evaluation with fault diagnosis using algorithmic fairness metrics (like Disparate Impact amp; Equal Opportunity) ensures that our RS's decision-making process is transparent and fair based on current labor laws.

IX. Conclusion and Future Work

Moving from hand-checking documents to smart systems powered by natural language processing marks a turning point in today's HR tools. This research introduces a finely tuned computational method that fixes core flaws in old-style keyword-focused applicant tracking. Instead of matching terms, it grasps meaning - thanks to advanced models like Sentence-BERT working alongside tailored entity detection. What once stopped at surface-level searches now digs into deeper understanding of candidate profiles. Efficiency rises when machines read between the lines, not just scan words. Results from the test back up the suggested S-BERT method. Better rankings showed up when checked through scores - NDCG landed at 0.812, MRR reached 0.744. What stands out most? A bi-encoder setup shaped like twins lets things grow fast, handling applicant files in just milliseconds. This screening approach wraps up fully built. Less weight now falls on hiring staff thanks to smoother operations. By tuning how meaning flows across data points, it softens mental shortcuts people make. Each person gets seen clearly - not split apart - for what they've truly done at work. Next steps involve pulling in models that handle many types of inputs together, aiming to hand tailored notes to those not selected. Clarity grows. So does how folks feel applying online.

References

[1] K. L. Abhishek, M. Farhan, N. Fatima, M. Umar, and N. Vashishth, "Developing an Intelligent Resume Screening Tool With AI-Driven Analysis and Recommendation Features," *Applied AI Letters*, 2025.

[2] N. Akhtar *et al.*, "AI-Driven Intelligent Resume Recommendation Engine," *International Journal of Scientific Engineering and Technology*, vol. 12, no. 3, 2025.

[3] C. Daryani *et al.*, "An Automated Resume Screening System Using Natural Language Processing and Similarity Measures," *International Conference on Intelligent Data Systems*, 2020.

[4] X. Q. Ong and K. H. Lim, "SkillRec: A Data-Driven Approach to Job Skill Recommendation," *arXiv preprint arXiv:2302.09938*, 2023.

[5] R. V. K. Bevara *et al.*, "Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings," *Electronics*, vol. 14, 2025.

[6] V. Arora and S. Maggu, "RAG-Enhanced LLM Job Recommendation Systems," *International Journal of Engineering Research and Technology*, vol. 14, 2025.

[7] "AI-Powered Resume Screening and HR Software," OrangeHRM, [Online]. Available: https://www.orangehrm.com. [Accessed: April 10, 2026].

[8] "AI Recruitment Trends 2026," Humanly, [Online]. Available: https://www.humanly.io. [Accessed: April 12, 2026].

[9] M. J. Smith and R. Taylor, "Overcoming Keyword Bias in Automated Applicant Tracking," *IEEE Transactions on Professional Communication*, vol. 63, no. 4, pp. 312-325, 2021.

[10] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171-4186.

[11] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982-3992.

[12] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed Representations of Words and Phrases and their Compositionality," in *Advances in Neural Information Processing Systems*, 2013, pp. 3111-3119.

[13] A. Vaswani *et al.*, "Attention Is All You Need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998-6008.

[14] M. Honnibal and I. Montani, "spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing," *To appear*, 2017.